Ex. No. 05	IMPLEMENTATION OF NAVIGATION AND ROUTING
Date of Exercise	
Git Hub Host Link	https://github.com/gurubruhh/Flutter-Laboratory.git

Aim

To implement navigation between multiple Flutter screens using navigation and routing techniques.

Description

This experiment demonstrates how to navigate between multiple screens in a Flutter application. Three screens, namely Home, Profile, and Details, are created. `Navigator.push()` and `MaterialPageRoute` are used for screen navigation, while `Navigator.pop()` is used to return to the previous screen. Named routes are also configured using the `routes` property of `MaterialApp`.

Procedure:

1. A new Flutter project named `navigation_app` was created using the Flutter command.
2. Three Dart files named `home.dart`, `profile.dart`, and `details.dart` were created inside the `lib` folder.
3. The Home screen was created with an "Go to Profile" button.
4. `Navigator.push()` with `MaterialPageRoute` was used to navigate from the Home screen to the Profile screen.
5. The Profile screen was created with a "View Details" button.
6. `Navigator.push()` was again used to navigate from the Profile screen to the Details screen.
7. `Navigator.pop()` was implemented in the Details screen to return to the previous screen.
8. Named routes were defined inside `MaterialApp` using the `routes` property for Home, Profile, and Details screens.
9. The application was executed using Flutter and tested in the Chrome browser.
10. The navigation flow was verified as Home → Profile → Details, with back navigation to the previous screen.

Program

Main.dart

```
import 'package:flutter/material.dart';
import 'home.dart';
import 'profile.dart';
import 'details.dart';


void main() {
  runApp(const NavigationApp());
}


class NavigationApp extends StatelessWidget {
  const NavigationApp({super.key});


  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Navigation App',
      initialRoute: '/',
      routes: {
        '/': (context) => const HomePage(),
        '/profile': (context) => const ProfilePage(),
        '/details': (context) => const DetailsPage(),
```

```
    },
   );
 }
```

home.dart

```
import 'package:flutter/material.dart';
import 'profile.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Home'),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
          Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) => const ProfilePage(),
```

```
            ),
          );
        },
        child: const Text('Go to Profile'),
      ),
    ),
  );
 }
}
```

profile.dart

```
import 'package:flutter/material.dart';
import 'details.dart';

class ProfilePage extends StatelessWidget {
 const ProfilePage({super.key});

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: const Text('Profile'),
   ),
   body: Center(
    child: ElevatedButton(
```

```
        onPressed: () {
          Navigator.push(
          context,
          MaterialPageRoute(
              builder: (context) => const DetailsPage(),
            ),
          );
        },
        child: const Text('View Details'),
      ),
    ),
  );
 }
}
```

details.dart

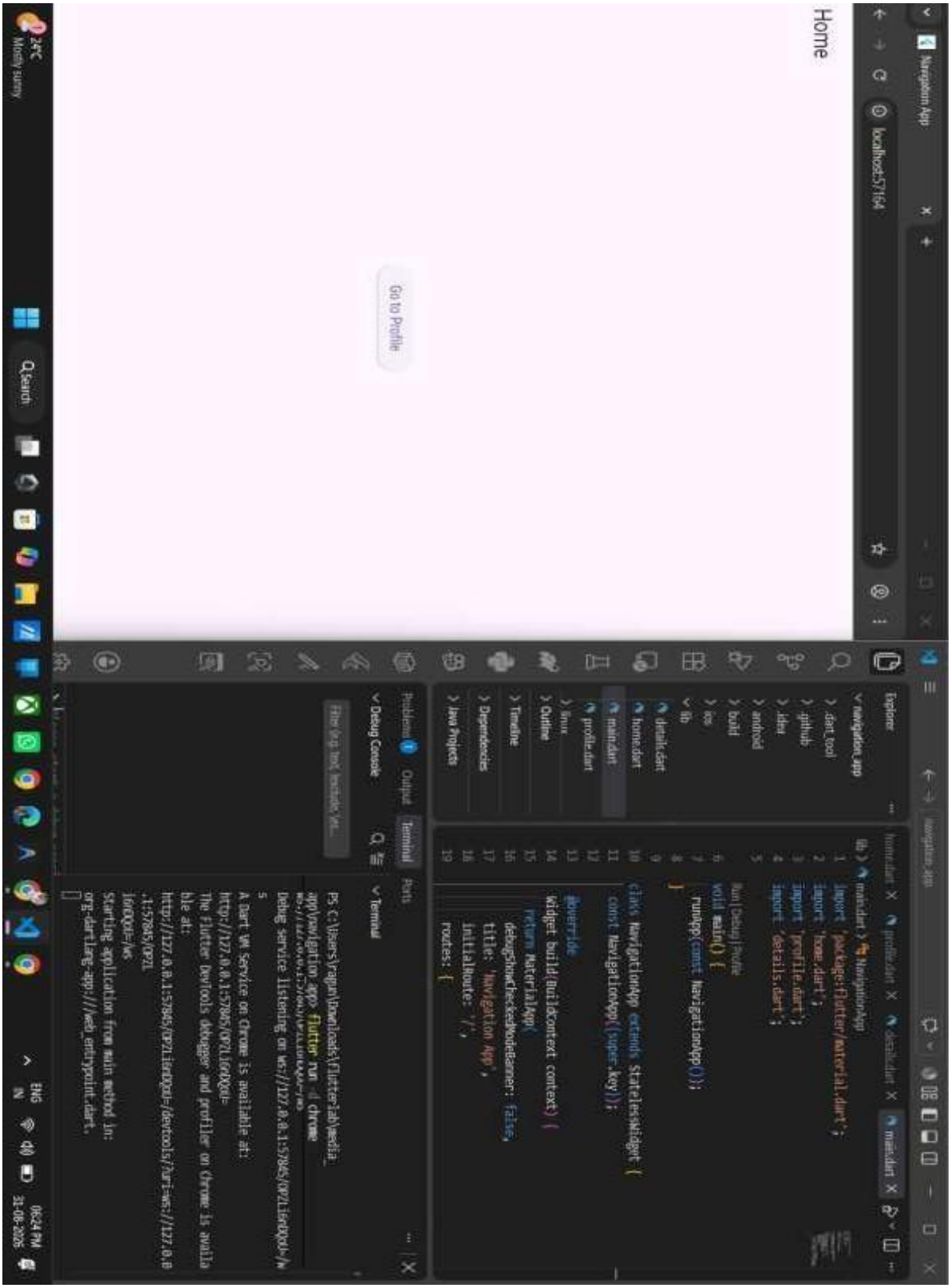
```
import 'package:flutter/material.dart';


class DetailsPage extends StatelessWidget {
  const DetailsPage({super.key});


  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

```
      appBar: AppBar(
        title: const Text('Details'),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
          Navigator.pop(context);
          },
          child: const Text('Back'),
        ),
      ),
    );
  }
}
```

Output



Result

Thus, the Flutter application was successfully developed to implement navigation and routing between multiple screens. The application successfully navigated from Home → Profile → Details and allowed the user to return to the previous screen using back navigation.